

c) The equation $Ax=b$ is consistent if the augmented matrix $[A \ b]$ has a pivot position in every row.

Solution: On Page 38, just above the section "Computation of Ax ", it is mentioned as a Warning that If an augmented matrix $[A \ b]$ has a pivot position in every row, then the equation $Ax=b$ may or may not be consistent.
 $\therefore$ The statement is False. $\square$

d) The first entry in the product Ax is a sum of products.

Solution: On Page 38, from the "Row-Vector Rule for Computing Ax ", the i th entry in Ax is the sum of the products of corresponding entries from row i of A and from the vector x .
 $\therefore$ The first entry in the product Ax is a sum of the products of corresponding entries from row 1 of A and from the vector x .
 $\therefore$ The statement is True, but it should be more precise. $\square$

e) If the columns of an $m \times n$ matrix A span $\mathbb{R}^m$, then the equation $Ax=b$ is consistent for each b in $\mathbb{R}^m$.

Solution: From Theorem 4 on Page 37, if the columns of A span $\mathbb{R}^m$, then for each b in $\mathbb{R}^m$, the equation $Ax=b$ has a solution. By defn, $\therefore$ the equation $Ax=b$ is consistent for each b in $\mathbb{R}^m$. $\therefore$ The given statement is True. $\square$

f) If A is an $m \times n$ matrix and if the equation $Ax=b$ is inconsistent for some b in $\mathbb{R}^m$, then A cannot have a pivot position in every row.

Solution: By Theorem 4 on Page 37, ~~if~~ A is an $m \times n$ matrix, for each b in $\mathbb{R}^m$, the equation $Ax=b$ has a solution if and only if A has a pivot position in every row. $\therefore$ If $Ax=b$ does not have a solution for some b in $\mathbb{R}^m$, i.e. $Ax=b$ is inconsistent for some b in $\mathbb{R}^m$ (by defn. of inconsistency), then A cannot have a pivot position in every row.

24.a) Every matrix equation $Ax=b$ corresponds to a vector equation with the same solution set.